q4_grades.c - Code::Blocks / VS Code (C)

```c
#include <stdio.h>

char grade_ladder(int m)
{
    if (m < 0 || m > 100) return '?';
    if (m >= 90) return 'S';
    else if (m >= 80) return 'A';
    else if (m >= 70) return 'B';
    else if (m >= 60) return 'C';
    else if (m >= 50) return 'D';
    else return 'F';
}

char grade(int marks)
{
    if (marks < 0 || marks > 100) return '?';
    switch (marks / 10) {
        case 10:
        case 9:  return 'S';
        case 8:  return 'A';
        case 7:  return 'B';
        case 6:  return 'C';
        case 5:  return 'D';
        default: return 'F';
    }
}

char grade_table(int m)
{
    static const char grades[11] = {'F','F','F','F','F','D','C','B','A','S','S'};
    if (m < 0 || m > 100) return '?';
    return grades[m / 10];
}

const char *scholarship(char g, int income)
{
    static const char *tab[6][3] = {
        {"Full", "Full", "Half"}, {"Full", "Half", "Quarter"}, {"Half", "Quarter", "None"},
        {"Quarter", "None", "None"}, {"None", "None", "None"}, {"None", "None", "None"}};
    const char *order = "SABCDF";
    int row = -1;
    for (int i = 0; i < 6; i++) if (order[i] == g) row = i;
    if (row < 0) return "Invalid";
    int band = income < 300000 ? 0 : income < 800000 ? 1 : 2;
    return tab[row][band];
}

int main(void)
{
    int marks[] = {100, 95, 83, 50, 49, -5};
    for (int i = 0; i < 6; i++) printf("%d:%c ", marks[i], grade(marks[i]));
    printf("\n");

    int tests[] = {100, 90, 89, 80, 79, 70, 69, 60, 59, 50, 49, 0, -1, 101};
    printf("marks ladder switch table\n");
    for (int i = 0; i < 14; i++)
        printf("%5d   %c      %c      %c\n", tests[i], grade_ladder(tests[i]),
               grade(tests[i]), grade_table(tests[i]));

    printf("Scholarship (A, income 250000) = %s\n", scholarship('A', 250000));
    printf("Scholarship (C, income 900000) = %s\n", scholarship('C', 900000));

    return 0;
}
```

Terminal - output

```
user@pc:~/Assignment2$ gcc -Wall q4_grades.c -o q4 && ./q4
100:S 95:S 83:A 50:D 49:F -5:?
marks ladder switch table
  100   S      S      S
   90   S      S      S
   89   A      A      A
   80   A      A      A
   79   B      B      B
   70   B      B      B
   69   C      C      C
   60   C      C      C
   59   D      D      D
   50   D      D      D
   49   F      F      F
    0   F      F      F
   -1   ?      ?      ?
  101   ?      ?      ?
Scholarship (A, income 250000) = Full
Scholarship (C, income 900000) = None
user@pc:~/Assignment2$ _
```